

Chronic Kidney Disease (CKD) Classification – Paul Immanuel

The aim is to develop a machine learning model that can predict whether a patient is affected by Chronic Kidney Disease (CKD) based on several medical parameters. The dataset was provided to build an accurate predictive model for disease classification.

Dataset info:

Dataset Name: CKD.csv

Number of Rows: 400

Number of Columns: 25

Target Variable: Classification

The dataset contains various medical attributes such as age, blood pressure, blood glucose level, blood urea, serum creatinine, sodium, potassium, haemoglobin and other health-related factors that are useful in identifying Chronic Kidney Disease.

Preprocessing:

- Loaded the dataset using Pandas.
- Converted categorical variables into numerical format using `pd.get_dummies()`.
- Used `drop_first=True` to avoid dummy variable redundancy.
- Split the dataset into training and testing datasets using Train-Test Split.
- Training data: 67%
- Testing data: 33%

ML Models used:

1. Support Vector Machine (SVM)
2. Random Forest
3. Decision Tree
4. Logistic Regression
5. K-Nearest Neighbors (KNN)
6. Gaussian Naive Bayes
7. Multinomial Naive Bayes
8. Bernoulli Naive Bayes
9. Complement Naive Bayes

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Model evaluation results

S.No	Algorithm	Accuracy
1	Random Forest	98.50%
2	GaussianNB	97.74%
3	BernoulliNB	93.98%
4	Logistic Regression	93.98%
5	Decision Tree	92.48%
6	ComplementNB	81.95%
7	MultinomialNB	81.95%
8	KNN	69.17%
9	SVM	61.65%

Selected Model: Random Forest Classifier, Accuracy: 98.50%

Random Forest produced the highest accuracy among all the machine learning algorithms tested. It outperformed Gaussian Naive Bayes, Logistic Regression, Decision Tree, KNN and SVM models.

Advantages of Random Forest:

- Highest prediction accuracy.
- Handles large datasets effectively.
- Reduces overfitting through ensemble learning.
- Provides robust and reliable classification performance.

Therefore, Random Forest was selected as the final model for predicting Chronic Kidney Disease.

Screenshots:

Random Forest classification report:

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Random Forest					
[[50 1]					
[1 81]]					
	precision	recall	f1-score	support	
False	0.98	0.98	0.98	51	
True	0.99	0.99	0.99	82	
accuracy			0.98	133	
macro avg	0.98	0.98	0.98	133	
weighted avg	0.98	0.98	0.98	133	
Accuracy : 0.9849624060150376					

Final accuracy comparison table:

OUTPUT		
<pre>result_df = pd.DataFrame(results, columns=["Algorithm", "Accuracy"]) result_df.sort_values(by="Accuracy", ascending=False)</pre>		
Algorithm	Accuracy	
1 Random Forest	0.984962	
5 GaussianNB	0.977444	
7 BernoulliNB	0.939850	
3 Logistic Regression	0.939850	
2 Decision Tree	0.924812	
8 ComplementNB	0.819549	
6 MultinomialNB	0.819549	
4 KNN	0.691729	
0 SVM	0.616541	

Conclusion

In this, multiple machine learning classification algorithms were applied to the CKD dataset. After preprocessing and evaluation, the Random Forest Classifier achieved the highest accuracy of 98.50%. Hence, it was chosen as the final predictive model for classification.